The Exact Log-Concavity Threshold for Partitions into Cubes

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24 October 2026

Abstract

Let $c(n)$ count unordered partitions of n into positive cubes, with unlimited repetitions. We prove that $c(n)^2 > c(n-1)c(n+1)$ for every $n \geq 15656$, whereas the difference is negative at 15655. This settles the cubic case of the conjectured sharp log-concavity thresholds for power partitions. We determine all 5505 negative indices and all 41 equality indices. The proof joins an exact integer calculation through ten million to an explicit analytic estimate for the infinite tail. A cubic Mellin transformation, a twenty-fourth-order saddle expansion with a proved remainder, and a finite rational covering of the noncentral arcs give an effective coefficient bound. Errors are bounded at the three coefficient arguments before taking a logarithmic second difference. All proof-critical computations are reproducible using short Python programs with exact integer and rational arithmetic.

1 The theorem and the sharp boundary

Define the cube-partition function by

$$\mathcal{F}(q) = \sum_{n\geq 0} c(n)q^n = \prod_{j\geq 1}(1-q^{j^3})^{-1}, \qquad c(0) = 1. \tag{1}$$

Thus order is ignored, repetitions are unlimited, and parts are positive perfect cubes. For example $c(9) = 2$, from nine ones and from $8+1$. This is different from partitions into distinct cubes or representations by a prescribed number of cubes. For $n \geq 1$ let

$$D_3(n) = c(n)^2 - c(n-1)c(n+1), \qquad T_3 = \min\{t \in \mathbb{Z}_{\geq 1} : D_3(n) \geq 0 \text{ for every integer } n \geq t\}.$$

Theorem 1.1. *The exact permanent nonnegative log-concavity threshold is $T_3 = 15656$. In fact $D_3(n) > 0$ for every integer $n \geq 15656$. The complete negative set has 5505 elements and is the disjoint union of the arithmetic progressions in Appendix A. The complete equality set is*

$$\begin{aligned}\{&1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, 17, 18, 19, 20, 21, 22, 25,\\ &28, 29, 30, 33, 36, 37, 38, 41, 44, 45, 46, 49, 52, 57, 60, 65, 68,\\ &73, 76, 84, 92, 100\}.\end{aligned}$$

At every other positive integer the difference is strictly positive.

The boundary values are

$$D_3(15655) = -434145527131309396, \qquad D_3(15656) = 85549986172343907934. \tag{2}$$

Finite values alone cannot prove permanence. The proof below supplies both the analytic infinite range and an exact adjoining finite range.

1.1 Prior work and the novelty boundary

Roy [1], Section 3.1, Conjecture 3.3 and Table 3, proposes $N_3 = 15655$ under the convention that log-concavity holds for $n > N_3$. Our convention is $n \geq T_3$, so the proposed value here is $T_3 = N_3 + 1 = 15656$. Both the version of record and arXiv version 4 [2] list Roy as sole author; earlier preprint versions have different authorship.

Eventual log-concavity was already known. Benfield's thesis [6] gives an eventual theorem and numerical boundary evidence. O'Sullivan [3], Theorem 1.3 and Corollary 5.7, proves eventual strengthened log-concavity and a second-difference expansion for power partitions. Benfield, Paul and Roy [4], Theorem 1.3 and Corollary 1.4, prove eventual Turán inequalities, with the smallest threshold left conjectural. These results do not supply the effective cubic constants and cutoff required to connect the sharp boundary to a finite calculation.

Ulas [5], Conjecture 3.5 and the discussion on p. 413, already reports finite verification of a strengthened cubic inequality on $15656 \leq n \leq 10^6 - 1$. OEIS A346160 [8] lists 15655 using the strict-after convention; its examples contain inconsistent off-by-one wording. We use the original theorem conventions and exact boundary arithmetic, rather than those examples.

A search on 24 October 2026 inspected these original statements and equivalent threshold and Turán formulations. No earlier complete proof of this exact cubic threshold was located. This is a documented search finding, not a guarantee about unindexed work. The contribution here is the effective infinite-range proof joined to the exact boundary and full exception classification. We make no novelty claim for eventual log-concavity or the counting recurrences.

The Mellin argument is a cubic specialization of the generalized eta transformation framework studied by Zagier [7]; the needed identity, branches and error bounds are derived here. The broad analytic and computational architecture parallels the author's companion treatment of partitions into squares. No result from that study is assumed, and all cubic specializations are proved in this paper.

1.2 Constants and coverage

Put

$$\begin{gathered} A = \Gamma(4/3)\zeta(4/3), \quad a = (A/3)^{3/4}, \quad m = n + 1/240, \\ t = (A/(3m))^{3/4}, \quad x = t^{-1/3}, \quad X = mt = (A/3)x = am^{1/4}. \end{gathered} \tag{3}$$

The shift $1/240$ is essential: it is the residue contribution at $s = -1$ in the cubic transformation. We use

$$321/100 < A < 161/50, \qquad 157/50 < \pi < 22/7, \qquad a > 1. \tag{4}$$

Appendix B proves and certifies these numerical bounds. Let $N_0 = 10{,}000{,}000$. For $n \geq N_0$,

$$m \geq N_0 > \frac{161}{150}50^4, \qquad x > 50, \qquad X > 50, \qquad t < 1/125000.$$

We prove a relative coefficient approximation of order X^{-7}. Since $X = am^{1/4}$, this is exactly the $n^{-7/4}$ scale needed for log-concavity. The leading logarithmic curvature of the model is $(3/4)am^{-7/4}$; its smaller terms are controlled explicitly.

2 The cubic transformation and explicit constants

Write $F(q) = \mathcal{F}(q)$ for the product in (1), and put

$$A = \Gamma(4/3)\zeta(4/3), \qquad K = (2\pi)^{4/3}, \qquad K_0 = K, \quad K_\pm = Ke^{\pm i\pi/3}.$$

All powers and logarithms below use their principal branches.

Lemma 2.1. *For* $\operatorname{Re} z > 0$,

$$\log F(e^{-z}) = Az^{-1/3} + \tfrac{1}{2}\log z - \tfrac{3}{2}\log(2\pi) + \frac{z}{240} + R(z), \tag{5}$$

where the following double series converges absolutely:

$$R(z) = \sum_{l,m\geq 1} \frac{1}{l} \sum_{\epsilon\in\{0,+,-\}} \exp\{-K_\epsilon l(m/z)^{1/3}\}. \tag{6}$$

The logarithm on the left is defined by the absolutely convergent logarithmic series of the Euler product.

Proof. Mellin inversion, initially on $\operatorname{Re} s = c > 1/3$, gives

$$\log F(e^{-z}) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \Gamma(s)\zeta(1+s)\zeta(3s)z^{-s}\,ds.$$

The two Dirichlet series converge absolutely there. Shift the contour to $\operatorname{Re} s = -3/2$. The crossed poles are $s = 1/3, 0, -1$. Their residues are respectively

$$Az^{-1/3}, \qquad \tfrac{1}{2}\log z - \tfrac{3}{2}\log(2\pi), \qquad z/240.$$

For the last residue, use $\operatorname{Res}_{s=-1}\Gamma(s) = -1$, $\zeta(0) = -1/2$, and $\zeta(-3) = 1/120$. At zero, $\Gamma(s)\zeta(1+s) = s^{-2} + O(1)$ and $\zeta'(0) = -\frac{1}{2}\log(2\pi)$. On the fixed strip the zeta factors have polynomial growth, whereas Stirling's formula gives exponential decay $e^{-\pi|\operatorname{Im} s|/2}$ from the gamma factor. Since $|\arg z| < \pi/2$, the factor z^{-s} preserves exponential decay and the horizontal connecting integrals vanish.

Set $w = -3s$ in the remaining integral. The functional equations

$$\begin{aligned}
\zeta(1-w/3) &= 2^{1-w/3}\pi^{-w/3}\cos(\pi w/6)\Gamma(w/3)\zeta(w/3),\\
\zeta(-w) &= -2^{-w}\pi^{-w-1}\sin(\pi w/2)\Gamma(1+w)\zeta(1+w),\\
\Gamma(-w/3)\Gamma(w/3) &= -\frac{3\pi}{w\sin(\pi w/3)}
\end{aligned}$$

give the meromorphic identity

$$\begin{aligned}
&\tfrac{1}{3}\Gamma(-w/3)\zeta(1-w/3)\zeta(-w)\\
&\quad= \Gamma(w)\zeta(1+w)\zeta(w/3)(2\pi)^{-4w/3}\frac{\sin(\pi w/2)}{\sin(\pi w/6)}\\
&\quad= \Gamma(w)\zeta(1+w)\zeta(w/3)(K_0^{-w} + K_+^{-w} + K_-^{-w}).
\end{aligned}$$

The factor $1/3$ and orientation reversal are included in this identity. On $\operatorname{Re} w = 9/2$, expand $\zeta(1+w)\zeta(w/3) = \sum_{l,m\geq 1} l^{-1-w}m^{-w/3}$. The absolute sum converges. Further,

$$\arg(K_\epsilon z^{-1/3}) \in (-\pi/2, \pi/2) \qquad (\epsilon = 0, +, -).$$

The exponential decay of the gamma factor, even after multiplication by the three phase factors and $z^{w/3}$, justifies interchanging the sum and integral. Mellin inversion now gives (6). Its absolute convergence also follows directly from the bound below. □

Lemma 2.2. *Put* $\theta = \arg z$ *and*

$$b(z) = K|z|^{-1/3}\cos(\pi/3 + |\theta|/3) > 0.$$

Then

$$|R(z)| \le E(b(z)), \qquad E(b) = \frac{3e^{-b}}{1-e^{-b}}\left(1 + \frac{3}{b} + \frac{6}{b^2} + \frac{6}{b^3}\right). \tag{7}$$

Proof. The number $b(z)$ is the least real part of $K_\epsilon z^{-1/3}$. Discard $1/l \le 1$, sum the geometric series in l, and use

$$\sum_{m\ge 1} e^{-bm^{1/3}} \le e^{-b} + \int_1^\infty e^{-bu^{1/3}}\,du = e^{-b}\left(1 + \frac{3}{b} + \frac{6}{b^2} + \frac{6}{b^3}\right).$$

This proves the claim. □

We will use the following rational bounds:

$$\frac{321}{100} < A < \frac{161}{50}, \qquad \frac{157}{50} < \pi < \frac{22}{7}. \tag{8}$$

Here is a reproducible derivation, so that no floating-point evaluation of the special functions is needed. Since $\Gamma(4/3) = \int_0^\infty e^{-u^3}\,du$, integrate the odd and even Taylor polynomials of e^{-u^3} of degrees 61 and 60 in u^3 over $[0,2]$. They give a lower and upper bound for that part of the integral. The remaining tail is at most $e^{-8}/12$, because $u^3 \ge 8 + 12(u-2)$ for $u \ge 2$. Bound e^8 below by its positive Taylor sum through degree 180. For $\zeta(4/3)$, sum the first 1000 terms, bounding each cube root between consecutive multiples of 10^{-12} by integer cubing, then use

$$\frac{3}{1001^{1/3}} < \sum_{j>1000} j^{-4/3} < \frac{3}{1000^{1/3}}.$$

All these calculations are rational and give (8). The bounds for π follow, also rationally, from Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ and the alternating series through indices 19 and 20. The supplied verifier implements exactly these calculations.

Proposition 2.3. *Use* $m = n + 1/240$, $t = (A/(3m))^{3/4}$, $x = t^{-1/3}$, *and* $X = mt = (A/3)x$, *and let* F_0 *and* M *be as in* (10). *For* $x \ge 50$ *and* $z = t + iy$ *with* $|y| \le 3t$,

$$b(z) > \tfrac{4}{5}x, \qquad |R(z)| \le 4e^{-4x/5}, \qquad |e^{R(z)} - 1| \le 8e^{-4x/5}.$$

On the positive real axis $e^{R(t)} < 101/100$. *Moreover,*

$$\frac{1}{M(m)}\left|\frac{1}{2\pi i}\int_{t-3it}^{t+3it} F(e^{-z})e^{nz}\,dz - V(m)\right| < \frac{1}{500}X^{-7}. \tag{9}$$

Proof. For $r \ge 0$, the function $\mathrm{Re}\{e^{i\pi/3}(1-ir)^{-1/3}\}$ decreases: its derivative is $-\frac{1}{3}|1-ir|^{-4/3}\sin(\pi/3 + 4\arctan(r)/3) < 0$. At $r = 3$, put $v = e^{i\pi/3}(1-3i)^{-1/3}$, $a_0 = \mathrm{Re}\,v > 0$, and $R_0 = |v|^2 = 10^{-1/3} < 47/100$. Since $v^3 = (-1-3i)/10$,

$$a_0(3R_0 - 4a_0^2) = 1/10, \qquad a_0 > \frac{1}{30R_0} > \frac{10}{141}.$$

Also $K > 23/2$, by (8) and integer cubing. Hence $b(z) > (115/141)x > (4/5)x$. For $b \ge 40$, (7) gives $E(b) < 4e^{-b}$; one may use $e^{-40} < 1/1000$ in its prefactor. The elementary inequality $e^u \le (1-u)^{-1}$ for $0 \le u < 1$ gives both the displayed exponential bound and $e^{R(t)} < 101/100$.

On the segment, $|z|^{1/2} \le 10^{1/4}\sqrt{t} < (9/5)\sqrt{t}$ and $\operatorname{Re} z^{-1/3} \le t^{-1/3}$. Inserting the segment length $6t$, the factor $1/(2\pi)$, and the preceding error bound, then dividing by M, bounds the left side of (9) by

$$24\,10^{1/4}\sqrt{\frac{8X}{3\pi}}e^{-4x/5} < 42\sqrt{x}\,e^{-4x/5}.$$

The term $z/240$ in (5) is absorbed exactly in $m = n + 1/240$ here. Since $X < (161/150)x$, multiplication by X^7 leaves a constant times $x^{15/2}e^{-4x/5}$, which decreases for $x \ge 50$. At $x = 50$, use $\sqrt{50} < 71/10$ and the positive degree-180 Taylor sum for e^{40}. Exact rational arithmetic gives

$$\frac{42(71/10)(161/3)^7}{\sum_{k=0}^{180} 40^k/k!} < \frac{1}{500}.$$

This proves the bound on the whole infinite domain. □

3 An explicit saddle approximation

Put $m = n + 1/240$, $a = (A/3)^{3/4}$, $t = (A/(3m))^{3/4}$ and $X = mt = am^{1/4}$, and set $C = (2\pi)^{-3/2}$. The model product and its leading coefficient approximation are

$$F_0(z) = C\sqrt{z}\,e^{Az^{-1/3}}, \qquad M(m) = Ct^{3/2}e^{4X}\sqrt{\frac{3}{8\pi X}} = \frac{\sqrt{3}a}{8\pi^2}m^{-5/4}e^{4am^{1/4}}. \tag{10}$$

All fractional powers use the principal branch. Define

$$V(m) = \frac{1}{2\pi i}\int_{t-3it}^{t+3it} F_0(z)e^{mz}\,dz.$$

This section proves an explicit approximation to V; the transformation and other arcs are treated separately.

3.1 The steepest contour

Let $H(m)$ be the same inverse Laplace integral on a Hankel contour around the negative real axis, staying a positive distance from zero. The contour may be deformed in the slit plane; no collapse onto the essential singularity at zero is used. Set $z = tw^3$. On the upper half of a steepest contour take

$$w = re^{i\phi}, \quad r^4 = \frac{3}{4\cos^2\phi - 1}, \quad 0 \le \phi < \pi/3, \qquad w^3 + 3/w = 4 - y^2. \tag{11}$$

The imaginary part vanishes because $r^3\sin(3\phi) = 3r^{-1}\sin\phi$. The real part is $4r^3c(2c^2-1)$, where $c = \cos\phi$; its derivative with respect to c is a positive factor times $12c^4 - 4c^2 + 1 > 0$. It therefore decreases from 4 to $-\infty$ as ϕ increases. Thus y parametrizes $[0,\infty)$ uniquely. At infinity the image z approaches the negative real axis from above with imaginary part tending to zero. The lower reflected path completes a Hankel contour. Deformations at large negative real part are justified by exponential decay. Combining the two halves gives

$$H(m) = \frac{3Ct^{3/2}e^{4X}}{\pi}\int_0^\infty e^{-Xy^2}g(y)\,dy, \qquad g(y) = \operatorname{Im}\{w(y)^{7/2}w'(y)\}. \tag{12}$$

Here $\sqrt{z}\,dz = 3t^{3/2}w^{7/2}\,dw$ and $g(0) = 1/\sqrt{6}$, which recovers exactly the normalization in (10).

Lemma 3.1. *The function g has an even holomorphic extension to $|y| < 6/5$. If $g(y) = \sum_{r\geq 0} g_{2r} y^{2r}$, then*

$$|g_{2r}| \leq 52(10/11)^{2r}, \qquad |g(y)| \leq 14 + 2y^2 \quad (y \geq 0).$$

For $0 \leq y \leq 1$,

$$\left| g(y) - \sum_{r=0}^{24} g_{2r} y^{2r} \right| \leq 300(10/11)^{50} y^{50}. \tag{13}$$

Proof. The algebraic equation for the selected branch is

$$(w-1)^2(w^2+2w+3) + y^2 w = 0, \qquad w(0) = 1, \quad w'(0) = i/\sqrt{6}. \tag{14}$$

On $|w-1| = 1$, put $w - 1 = e^{i\psi}$. Then $|w^2+2w+3|^2 = 24\cos^2\psi + 56\cos\psi + 41 \geq 9$. Also $|w| \leq 2$, and $2(6/5)^2 < 3$. Rouché's theorem puts exactly two roots inside $|w-1| < 1$ for $|y| < 6/5$. At zero they split analytically, as follows by substituting $w = 1 + yv$ and applying the implicit function theorem at $v = \pm i/\sqrt{6}$. The only other multiple-root parameters satisfy $y^2 = 8, 4+4i, 4-4i$, all outside this disk. Analytic continuation gives the selected branch throughout the disk, with no monodromy at zero. It is nonzero by (14), so its fractional powers are holomorphic, with the branch positive at zero.

Let $h = w^{7/2}w' = (2/9)(w^{9/2})'$. Since $|w| < 2$ and $2^{9/2} < 23$, Cauchy's derivative bound with the gap $6/5 - 11/10 = 1/10$ gives $|h| < 460/9 < 52$ on $|y| \leq 11/10$. The limiting-radius form of Cauchy's bound suffices at the endpoints. Define the holomorphic extension by $g = (h(y) + h(-y))/(2i)$. For real y, $w(-y) = \overline{w(y)}$ and $w'(-y) = -\overline{w'(y)}$, so this agrees with the original g. Cauchy's coefficient bound proves the assertion about g_{2r}, and summing the geometric tail gives (13), since $52/(1-(10/11)^2) < 300$.

For the global real bound, (11) gives $r \geq 1$ and $0 \leq \arg w < \pi/3$. Differentiating the phase and canceling $w - 1$ using (14) yields

$$|h| = \frac{2r^5|w^2+2w+3|^{1/2}}{3|(w+1)(w^2+1)|}.$$

The limiting value is used at zero. Here $|w+1| \geq r$, $|w^2+1| \geq \sqrt{3}r^2/2$, and $|w^2+2w+3|^{1/2} \leq \sqrt{6}r$. Hence $|h| < 2r^3$. The phase equation gives $r^3 \leq |4-y^2| + 3/r \leq 7 + y^2$, proving the bound on g. □

3.2 The rational expansion and its remainder

For $0 \leq r \leq 24$ define

$$c_r = \frac{(-1)^r(1/2)_r}{6^r}[z^{2r}](1+z)^{r+4}(1+2z/3+z^2/6)^{-r-1/2}, \qquad P(X) = \sum_{r=0}^{24} c_r X^{-r}. \tag{15}$$

Here $(b)_r$ is the rising factorial, with $(b)_0 = 1$. In particular, $c_0 = 1$ and $c_1 = -67/144$. For completeness, write $\delta = w - 1$ in (14) to obtain $\delta = iy\sqrt{(1+\delta)/(6+4\delta+\delta^2)}$. Lagrange inversion applied to $h = (2/9)(w^{9/2})'$ gives

$$g_{2r} = (-1)^r 6^{-r-1/2}[z^{2r}](1+z)^{r+4}(1+2z/3+z^2/6)^{-r-1/2}.$$

The Gaussian moment integral then gives $c_r = \sqrt{6}\, g_{2r}(1/2)_r$, proving (15) and its normalization.

Lemma 3.2. *For $X \geq 50$, with $B = 750(1/2)_{25}(10/11)^{50}$,*

$$|H/M - P| \leq BX^{-25} + 1000X^{-1/2}e^{-X}.$$

Proof. Integrate (13) on $[0,1]$, enlarge its upper bound to the half-line, and divide by $\sqrt{\pi}/(2\sqrt{6X})$. Since $\sqrt{6} < 5/2$, the local term is at most BX^{-25}. Both tails must also be bounded; no positivity of g is assumed. For $J_r = \int_1^\infty y^{2r} e^{-Xy^2}\,dy$, integration by parts gives

$$J_0 \le \frac{e^{-X}}{2X}, \qquad J_r = \frac{e^{-X}}{2X} + \frac{2r-1}{2X} J_{r-1}.$$

For $0 \le r \le 24$ and $X \ge 50$, this implies $J_r < e^{-X}/X$, as $47/(2X) \le 47/100 < 1/2$. The global bound gives $14J_0 + 2J_1 < 16e^{-X}/X$; the polynomial tail is at most $52\sum_{r=0}^{24}(10/11)^{2r} J_r < 300e^{-X}/X$. The normalizing factor is $2\sqrt{6X/\pi} < 3\sqrt{X}$, so 1000 is a valid upper constant for the combined tails. □

Lemma 3.3. *For $X \ge 50$, $|H - V|/M < 10X^{-1/2}e^{-9X/10}$.*

Proof. Deform the Hankel contour to the vertical segment together with the horizontal rays $z = t(v \pm 3i)$, $-\infty < v \le 1$. For an explicit check that the steepest path lies between these rays, put $u = \cos^2\phi$ in (11). Then $(\mathrm{Im}(z/t))^4 = 27(1-u)^2(4u-1) \le 27/4 < 3^4$ for $1/4 < u \le 1$. The maximum occurs at $u = 1/2$. Write $f(v) = \mathrm{Re}\{v + 3i + 3(v+3i)^{-1/3}\}$. For $v \le 0$, the real part of $(v+3i)^{-1/3}$ increases with v: its derivative is $-\mathrm{Re}(v+3i)^{-4/3}/3 > 0$. Its value at zero is $3^{-1/3}\sqrt{3}/2 < 2/3$. Thus $f(v) < v + 2$, and $|v+3i|^{1/2} \le 2 + |v|$. For $0 \le v \le 1$ we have $f'(v) \ge 1 - 3^{-4/3} > 3/4$, $f(1) < 1 + 3\cdot(7/10) = 31/10$, and $|v+3i|^{1/2} < 2$. After removing the common prefactor, either ray is bounded by

$$(2/X + 1/X^2)e^{2X} + \frac{8}{3X}e^{31X/10}.$$

Dividing the two-ray bound by M gives

$$\frac{|H-V|}{M} \le \sqrt{\frac{8}{3\pi X}}\left((2 + X^{-1})e^{-2X} + \frac{8}{3}e^{-9X/10}\right) < 10X^{-1/2}e^{-9X/10}.$$

The connections at negative infinity vanish because of e^{mz}. □

Proposition 3.4. *For every $X \ge 50$,*

$$|V/M - P| < \frac{1}{1000}X^{-7}. \tag{16}$$

Proof. Multiply the preceding bounds by X^7. Every resulting term decreases on $X \ge 50$: they are BX^{-18} and constant multiples of $X^{13/2}e^{-cX}$ for $c = 1, 9/10$. At 50 replace $X^{-1/2}$ by 1 for an upper bound. Exact rational arithmetic, using the degree-500 positive Taylor sum as a lower bound for the exponential, gives respectively upper bounds $1/100000, 1/100000, 1/10000$. Their sum is below $1/1000$. The short verifier `code/major_bounds.py` performs these rational comparisons and regenerates all coefficients in (15). □

4 Curvature of the explicit approximation

Let $Q(m) = M(m)P(am^{1/4})$.

Proposition 4.1. *For $X \ge 50$,*

$$P > 99/100, \quad X|P'| < 1/100, \quad X^2|P''| < 1/50, \qquad -(\log Q(m))'' \ge \frac{7}{10}am^{-7/4}.$$

Consequently, if $a(m-1)^{1/4} \ge 50$,

$$2\log Q(m) - \log Q(m-1) - \log Q(m+1) \ge \frac{7}{10}a(m+1)^{-7/4}. \tag{17}$$

Proof. Define $S_j = \sum_{r=1}^{24} |c_r| b_j(r) 50^{-r}$ with $b_0(r) = 1$, $b_1(r) = r$, $b_2(r) = r(r+1)$. Exact evaluation of (15) gives $S_0, S_1 < 1/100$ and $S_2 < 1/50$. These sums dominate the corresponding quantities for every $X \geq 50$. For $L = \log P$ and $X' = X/(4m)$,

$$m^2 \frac{d^2}{dm^2} L(X(m)) = \frac{X^2 L'' - 3XL'}{16}.$$

Its absolute value is bounded by

$$\frac{1}{16}\left(\frac{S_2}{1-S_0} + \frac{S_1^2}{(1-S_0)^2} + \frac{3S_1}{1-S_0}\right) < \frac{1}{100}.$$

All these are rational checks in `major_bounds.py`. Since $(\log M)'' = (-3X/4 + 5/4)/m^2$ and $3X/4 - 5/4 - 1/100 \geq 7X/10$ for $X \geq 50$, the derivative bound follows. Finally apply the exact identity

$$2f(m) - f(m-1) - f(m+1) = -\int_{-1}^{1} (1 - |u|) f''(m+u)\, du$$

to $f = \log Q$. Only the explicit polynomial approximation has been differentiated; the coefficient error will be bounded separately at the three integer arguments. □

5 Uniform control of the remaining arcs

For $t > 0$ define

$$\rho(t, \theta) = \frac{|F(e^{-t+i\theta})|}{F(e^{-t})}.$$

Taking the modulus of each Euler factor gives the exact formula

$$\rho(t, \theta) = \prod_{j \geq 1} \left(1 + \frac{\sin^2(j^3\theta/2)}{\sinh^2(j^3 t/2)}\right)^{-1/2}. \tag{18}$$

Every factor is at most one, so any finite subset bounds the product from above.

Lemma 5.1. *For $0 < t \leq 1/125000$ and $x = t^{-1/3}$,*

$$\int_{3t}^{\pi} \rho(t, \theta)\, d\theta \leq \frac{51t}{x} e^{-37x/50} + \frac{1}{1000}(e^{-3x/4} + (1100t)^9) + \pi 10^{-24}(125000t)^8. \tag{19}$$

Proof. When $j^3\theta \leq 1$ and $3t \leq \theta$, the inequalities $\sin u/u \geq 23/24$ and $\sinh u/u \leq 261/250$ on $[0, 1/2]$ give

$$\rho(t, \theta) \leq [1 + c(\theta/t)^2]^{-\lfloor \theta^{-1/3} \rfloor/2}, \qquad c = 21/25. \tag{20}$$

For example, the nonconstant tail of $\sinh u/u$ is at most $(1/24)/(1 - 1/80) = 10/237 < 11/250$, and $((23/24)/(261/250))^2 > 21/25$. Put $r = \theta/t$, $h(r) = r^{-1/3} \log(1 + cr^2)$ and $v = cr^2$. Then

$$h'(r) = \frac{6v/(1+v) - \log(1+v)}{3r^{4/3}}.$$

On $3 \leq r \leq 4$, the numerator decreases, since its derivative with respect to v is $(5 - v)/(1 + v)^2 < 0$. Its value at the right endpoint exceeds $2016/361 - 27/10 > 72/25$. Since $4^{4/3} < 32/5$, we have $h'(r) > 3/20$ throughout this interval. Furthermore,

$$\tfrac{1}{2} h(3) > 37/50, \qquad \sqrt{1 + cr^2} \leq 19/5.$$

The first assertion follows from $3^{1/3} < 29/20$ and $\log(214/25) > 1073/500$. Using $\lfloor \theta^{-1/3} \rfloor \geq \theta^{-1/3} - 1$ in (20), and integrating an exponential, gives

$$\int_{3t}^{4t} \rho(t,\theta)\, d\theta \leq \frac{19t}{5} e^{-37x/50} \int_3^\infty e^{-3x(r-3)/40}\, dr < \frac{51t}{x} e^{-37x/50}.$$

For $4t \leq \theta \leq 1/1000$, $\lfloor \theta^{-1/3} \rfloor \geq (9/10)\theta^{-1/3}$. For $r \geq 4$, the numerator of h' strictly decreases to negative infinity, so h has at most one turning point, a maximum. Its minimum on the interval is therefore attained at an endpoint. At $r = 4$, $(9/20)h(4) > 3/4$, because $4^{1/3} < 8/5$ and $\log(361/25) > 8/3$. At $\theta = 1/1000$, the exponent in the product bound is $(9/2)\log(1 + c/(10^6 t^2))$. Since $1000/\sqrt{c} < 1100$, this proves

$$\int_{4t}^{1/1000} \rho(t,\theta)\, d\theta \leq \frac{1}{1000}(e^{-3x/4} + (1100t)^9).$$

The logarithmic comparisons used above are verified rationally by upper and lower Taylor bounds for the exponential; no numerical logarithm is needed.

For the far arcs put $\alpha = \theta/(2\pi)$. If $1/1000 \leq \theta \leq \pi$, then $\alpha > 7/44000 > 159/10^6$. Cover the larger interval $[159/10^6, 1/2]$ by the 499841 closed intervals

$$I_k = [k/B, (k+1)/B], \qquad B = 10^6, \quad 159 \leq k < 500000.$$

For $1 \leq j \leq 50$ compute the rational numbers

$$d_{j,k} = \min_{\alpha \in I_k} \|j^3 \alpha\|, \qquad a_{j,k} = \left(\frac{250000 d_{j,k}}{j^3}\right)^2.$$

If $j^3 I_k$ contains an integer then $d_{j,k} = 0$; otherwise it is the smaller endpoint distance from an integer. The supplied exact integer calculation establishes, for every k,

$$\prod_{j=1}^{50} (1 + \lfloor a_{j,k} \rfloor) \geq 10^{48}, \qquad \#\{j : a_{j,k} \geq 1\} \geq 16. \tag{21}$$

The minimum product and minimum count found are respectively

$$14872100691592956718938800018641871541521940480000,$$
$$18,$$

both at $k = 125000$. The algorithm is small enough to specify fully: for each j, k set $L = j^3 k$, $U = j^3(k+1)$. If $\lfloor L/B \rfloor \neq \lfloor U/B \rfloor$, use $d = 0$; otherwise use $d = \min(L \bmod B, B - (U \bmod B))/B$. Then form the squared rational a and the displayed integer product and count. Endpoint coincidences give $d = 0$ by the same rule.

To relate the finite certificate to the full continuum, set $t_0 = 1/125000$. For $j \leq 50$, $j^3 t_0/2 \leq 1/2$ and thus $\sinh(j^3 t_0/2) \leq j^3 t_0$. The inequality $|\sin(\pi u)| \geq 2\|u\|$ shows that $a_{j,k}$ is a lower bound for the squared ratio in (18) when $t = t_0$ and $\alpha \in I_k$. By convexity, $\sinh(j^3 t/2) \leq (t/t_0)\sinh(j^3 t_0/2)$ for $t \leq t_0$. For $U_0 = (t_0/t)^2 \geq 1$ and $a \geq 0$,

$$\log(1 + U_0 a) - \log(1 + a) \geq \frac{a}{1+a} \log U_0,$$

as follows by integration in $\log U_0$. Since $\sum_j a_{j,k}/(1 + a_{j,k}) \geq 8$, (21) therefore gives the uniform bound

$$\rho(t,\theta) \leq 10^{-24}(125000t)^8 \quad (0 < t \leq t_0,\ 1/1000 \leq \theta \leq \pi). \tag{22}$$

Integrating over a set of length at most π completes the proof. □

Proposition 5.2. *For $x \geq 50$, the contribution of the arcs $3t \leq |\theta| \leq \pi$ to the coefficient $c(n)$, divided by $M(m)$, has absolute value less than $X^{-7}/16$.*

Proof. Cauchy's formula and conjugation symmetry bound its modulus by $e^{nt}F(e^{-t})\pi^{-1}\int_{3t}^{\pi}\rho(t,\theta)\,d\theta$. The exact shift $m = n + 1/240$ and (10) give

$$\frac{e^{nt}F(e^{-t})}{\pi M(m)} = e^{R(t)}\frac{\sqrt{8X/(3\pi)}}{t} < \frac{\sqrt{x}}{t}.$$

Indeed, the square of the constant needed for this inequality is at most $(101/100)^2(1288/1413) < 1$. Insert (19) and multiply by X^7. Since $X < (161/150)x$, it suffices to bound the sum

$$\left(\frac{161}{150}\right)^7\left(51x^{13/2}e^{-37x/50} + \frac{x^{21/2}}{1000}e^{-3x/4} + \frac{1100^9}{1000}x^{-33/2} + \frac{22}{7}10^{-24}125000^8x^{-27/2}\right). \tag{23}$$

Each term decreases for $x \geq 50$. For the exponential terms this follows by logarithmic differentiation; the other exponents are negative. At $x = 50$, replace square roots by $7 < \sqrt{50} < 71/10$ and exponentials below by their positive Taylor sums through degree 180. Exact rational arithmetic bounds the sum by $0.060056 < 1/16$. This proves the claimed bound on the entire unbounded range. □

6 From coefficient bounds to the infinite tail

The estimates of the preceding sections give, for every integer $n \geq N_0$,

$$\left|\frac{c(n)}{M(n+1/240)} - P\big(a(n+1/240)^{1/4}\big)\right| < \frac{1}{10}X^{-7}. \tag{24}$$

Indeed Cauchy's coefficient formula is

$$c(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\mathcal{F}(e^{-(t+i\theta)})e^{n(t+i\theta)}\,d\theta.$$

Split at $|\theta| = 3t$. The transformation bounds replacement on the central segment by $MX^{-7}/500$. Proposition 3.4 bounds the model error by $MX^{-7}/1000$. The minor-arc bound is $MX^{-7}/16$. Their sum is $131/2000 < 1/10$. All three estimates hold on $x \geq 50$, $X \geq 50$, hence on $n \geq N_0$.

Since $P > 99/100$, (24) implies

$$\left|\frac{c(n)}{Q(m)} - 1\right| < \frac{10}{99}X^{-7} < \frac{1}{11}.$$

The elementary bound $|\log(1+u)| \leq |u|/(1-|u|)$ therefore gives

$$|\log c(n) - \log Q(m)| < \frac{1}{9}a^{-7}m^{-7/4} \quad (n \geq N_0). \tag{25}$$

Proposition 6.1. *$D_3(n) > 0$ for every integer $n \geq N_0 + 1$.*

Proof. At such an index the coefficient bound applies at $n-1, n, n+1$. The total error in their logarithmic second difference is less than $(4/9)a^{-7}(m-1)^{-7/4}$, whereas (17) gives a model second difference at least $(7/10)a(m+1)^{-7/4}$. For $m \geq 100$, Bernoulli's inequality gives

$$\left(\frac{m-1}{m+1}\right)^{7/4} \geq 1 - \frac{7}{2(m+1)} > \frac{9}{10}.$$

Since $a^8 = (A/3)^6 > 1$,

$$\frac{7}{10}a^8\left(\frac{m-1}{m+1}\right)^{7/4} > \frac{63}{100} > \frac{4}{9}.$$

Thus $2\log c(n) - \log c(n-1) - \log c(n+1) > 0$. The all-ones partition ensures $c(n) > 0$, so exponentiation proves the assertion. The error in (24) was never differentiated; its three absolute values were bounded directly. □

7 Exact computation and minimality

To cover $D_3(n)$ for $1 \le n \le N_0$, coefficients are needed through $B = N_0 + 1 = 10{,}000{,}001$. Parts greater than B cannot affect these coefficients. Let $C_j(k)$ count partitions of k with allowed parts $1^3, \dots, j^3$, with $C_j(k) = 0$ for $k < 0$. Separating partitions according to whether they contain j^3 gives

$$C_j(k) = C_{j-1}(k) + C_j(k - j^3), \quad C_0(0) = 1, \quad C_0(k) = 0 \ (k > 0). \tag{26}$$

Removing one j^3 is a bijection for the second term, so no partition is omitted or counted with an ordering. The ascending update

```
values = [1] + [0]*B
j = 1
while j**3 <= B:
    cube = j**3
    for k in range(cube, B+1):
        values[k] += values[k-cube]
    j += 1
```

implements (26): before a pass the array holds C_{j-1}; during the pass the smaller indexed entry already holds C_j. Induction proves that the resulting integers equal $c(k)$ for every $0 \le k \le B$. Python integers have arbitrary precision.

7.1 A genuinely different cross-check

Formal logarithmic differentiation of (1) yields

$$\frac{q\mathcal{F}'(q)}{\mathcal{F}(q)} = \sum_{r \ge 1} b(r)q^r, \qquad b(r) = \sum_{j^3 \mid r} j^3.$$

Each formal coefficient involves finitely many terms, so no analytic convergence assumption is needed. Comparing coefficients gives

$$nc(n) = \sum_{r=1}^{n} b(r)c(n-r), \qquad c(0) = 1. \tag{27}$$

This determines the coefficients uniquely by induction. A distinct implementation constructs b on multiples of cubes and computes this recurrence, checking every division for zero remainder. It agrees with coin change through coefficient 15657, including the entire exceptional region and both boundary determinants. The separate verifier also checks every identity (27) on this prefix. Its quadratic cost is why this different-algorithm check is confined to that explicit range. The full range through B was independently regenerated by another implementation of the product algorithm; it is not described as a full-range check by the divisor recurrence.

7.2 The finite result

The complete integer calculation and separate regeneration give 5505 negatives, 41 zeros, and 9,994,454 positives on $1 \le n \le 10{,}000{,}000$. The negative set is listed in Appendix A; the equality set is the one in Theorem 1.1. The largest negative is 15655, the largest zero is 100, and every index from 15656 through 10,000,000 is strictly positive. The four boundary coefficients are

n	$c(n)$
15654	37343677726723
15655	37368837361134
15656	37394013958024
15657	37419205227863

Substituting them in the definition of D_3 gives (2). The compact certificate records the exact sets and values. Its verifier reconstructs every coefficient and determinant from scratch. Hashes identify the canonical sequence and source artifacts; they do not replace the recurrences or exact sign checks.

Argument	Coefficients available	Determinants covered
Exact product computation	$0 \le k \le 10{,}000{,}001$	$1 \le n \le 10{,}000{,}000$
Divisor cross-check	$0 \le k \le 15657$	$1 \le n \le 15656$
Analytic coefficient bound	$k \ge 10{,}000{,}000$	$n \ge 10{,}000{,}001$

Table 1: Inclusive coverage. The divisor check is additional validation, not a replacement for the full finite calculation.

Proof of Theorem 1.1. The finite result and Proposition 6.1 cover all positive integers without a gap, as Table 1 shows. They give precisely the negative and equality sets in the theorem, and strict positivity elsewhere. In particular $D_3(n) > 0$ for all $n \ge 15656$, while $D_3(15655) < 0$. The definition of T_3 proves both the claimed threshold and its minimality. □

8 Verification scope

There are two computational roles. The finite partition calculation uses exact integers and proved counting recurrences. The analytic certificates use exact rational inequalities and a finite covering by closed intervals to prove bounds over continuous domains. Floating-point values in diagnostic reports are informational; no sign or proof-critical comparison depends on them. The infinite domain beyond a certified endpoint is covered by proved monotonicity, not sampled numerically.

The proof, implementations and manuscript were developed by GPT-6 Astra, an AI system, which also executed the checks. Separate internal reviews examined the saddle argument, transformation, minor arcs, index coverage and computation. This is internal adversarial review, not independent human peer review. No Lean or other proof-assistant theorem was compiled; no formal-verification claim is made. The supplementary audit records the checks, corrections and final review scope. The complete package was tested by extracting it into a fresh directory and running its documented verification commands.

A The complete negative set

For $a \le b$ with $a \equiv b \pmod 8$, write $[a,b]_8 = \{a, a+8, \ldots, b\}$. The negative set is the disjoint union of the 118 progressions below. A row with $a = b$ denotes a singleton. Each pair of columns is read independently; the two progressions on a row have no special relationship. Expansion gives exactly 5505 indices. Both this table and the full individual list are checked by the supplied programs against the exact coefficient calculation.

a	b	Count	a	b	Count
7	15527	1941	1473	1473	1
26	12682	1583	1497	1497	1
53	181	17	1537	1561	4
188	228	6	1581	1581	1
205	245	6	1625	1625	1
252	316	9	1662	1662	1
269	397	17	1681	1713	5
340	13492	1645	1726	1726	1
421	421	1	1737	1737	1
437	437	1	1750	1750	1
441	441	1	1753	1785	5
461	461	1	1841	1841	1
465	481	3	1865	1865	1
485	485	1	1878	1878	1
501	501	1	1889	1929	6
505	505	1	1902	1902	1
525	525	1	1950	1966	3
529	545	3	1953	2001	7
549	549	1	2017	2017	1
561	569	2	2041	2057	3
585	721	18	2078	2078	1
589	589	1	2081	2081	1
670	670	1	2102	2118	3
737	785	7	2105	2145	6
782	782	1	2142	2142	1
798	822	4	2166	2182	3
801	937	18	2169	2169	1
838	846	2	2193	2209	3
862	886	4	2345	2345	1
902	910	2	2409	2409	1
934	934	1	2453	2453	1
950	950	1	2517	2517	1
953	1001	7	2537	2537	1
966	974	2	2561	2561	1
990	998	2	2601	2601	1
1014	1038	4	2625	2625	1
1017	1089	10	2641	2641	1

Continued on the next page

a	b	Count	a	b	Count
1054	1062	2	2753	2753	1
1078	1126	7	2769	2777	2
1105	1153	7	2793	2793	1
1142	1150	2	2833	2841	2
1166	1166	1	2857	2857	1
1169	1177	2	2985	2993	2
1193	1193	1	3009	3009	1
1209	1217	2	3049	3057	2
1214	1214	1	3073	3073	1
1230	1238	2	3201	3201	1
1233	1241	2	12706	12746	6
1254	1254	1	12770	12810	6
1257	1281	4	12850	12874	4
1294	1294	1	12938	12938	1
1318	1318	1	13010	13010	1
1321	1329	2	13508	13556	7
1345	1345	1	13580	13620	6
1365	1365	1	13660	13684	4
1389	1389	1	13748	13748	1
1409	1409	1	13820	13820	1
1429	1429	1	15551	15591	6
1453	1453	1	15631	15655	4

B Certificates and portable reproduction

The accompanying archive contains the complete TeX sources, proof-critical programs, finite certificates, interval certificates, literature audit, review records and this PDF. It uses relative paths and requires no external files. The essential mathematical checks require Python 3.9 or later and its standard library only. Run from the extracted package root:

```
python code/verify_all.py
```

This command performs the full finite regeneration, divisor-prefix check, rational saddle estimates, rational special-function bounds, all 499841 closed-interval minor-arc checks, exception-table comparison and exact coverage checks. It writes a new `certificates/reproduction_report.json`. Runtime and interpreter metadata may differ on another machine; all mathematical comparisons and counts must pass. The documented run took minutes rather than hours in the recorded execution environment, but no portable timing guarantee is assumed. Optimization mode `-O` is rejected for scripts whose proof checks use assertions.

The individual commands are

```
python code/finite_verify.py certificates/finite_10000000.json
python code/major_bounds.py
python code/analytic_arcs.py
python code/exception_table.py
```

The finite verifier imports no generator code. It recomputes every coefficient through 10,000,001, checks every determinant through 10,000,000, and checks (27) through 15657. The canonical

coefficient encoding is one decimal integer per line, starting with $c(0)$, with ASCII newlines. Its SHA256 identifier is

`fd131d7c8527900fe60da6f611f2b48b3a6d420edc8c6c7189a9b31ceaffaf58.`

The compact certificate omits a bulky full coefficient file because the verifier regenerates it. It includes every negative and equality index and all boundary coefficients. The smaller, preserved 200000 run also includes its compressed full coefficient table.

The rational verifier for A and π implements the integral and series bounds proved in Section 2. For an exponential lower bound it uses $\sum_{k=0}^{d} q^k/k!$. When an upper bound is needed, it adds the first omitted term divided by $1 - q/(d+2)$, provided $0 \le q < d+2$; the remaining term ratios are then no larger than $q/(d+2)$. All comparisons use exact fractions. The far-arc verifier computes the minimum distance on every closed interval exactly, so its finite output implies the continuum bound (22). The monotonicity arguments in the text extend the certified endpoint estimates to every $x \ge 50$ and $X \ge 50$.

For clarity, the polynomial coefficients may be generated without any symbolic algebra library. In (15), expand

$$(1 + 2z/3 + z^2/6)^{-r-1/2} = \sum_{k \ge 0} \binom{-r-1/2}{k} \sum_{b=0}^{k} \binom{k}{b} (2/3)^{k-b} (1/6)^b z^{k+b}.$$

Only $k + b \le 2r$ contributes; multiplication by $(1+z)^{r+4}$ gives a finite rational sum. This is exactly what `major_bounds.py` evaluates. The review additionally compared all 25 coefficients against a distinct formal-series recurrence; that diagnostic comparison is not needed to assume the formula's validity.

To regenerate the main finite certificate into a new directory, run

```
python code/finite_generate.py --max-index 10000000 --omit-sequence --output-dir
    certificates/fresh
python code/finite_verify.py certificates/fresh/finite_10000000.json
```

The generator refuses to overwrite its previous results. A mid-run interruption may require restarting this short finite calculation; completed benchmark, intermediate-range and final certificates are retained.

The PDF is reproducible with pdfLaTeX and the packages declared in `paper/main.tex`. From the package root:

```
cd paper
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The resulting `paper/main.pdf` is the same manuscript as the distributed PDF; byte-for-byte PDF identity is not required, because TeX distributions and creation metadata may differ. The archive contains the essential supplemental proof and data. No external approval or formal proof-assistant result is asserted.

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